

SaveFood

Food Rescue and Relief Platform

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Status: Prototype / Portfolio Project

Platform: Mobile (iOS and Android)

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Note: *SaveFood is not a real product. It was built to demonstrate a zero to one product mindset using the concepts of crowdsourcing, IoT integration, and social impact. The prototype was created as a portfolio piece to show how a product manager thinks through a multi-sided platform from the ground up, without an engineering team. It is intended for hiring managers, mentors, and peers in the product management community.*

1. Overview

What is SaveFood?

SaveFood is a mobile app that connects restaurants with leftover food to the people and nonprofits who need it. Instead of throwing away unsold food at the end of the day, a restaurant can post it on the app and a nearby nonprofit or community member can come pick it up before it expires.

The app also lets donors send money directly to nonprofits so they can keep running their food rescue operations. Every dollar goes to the nonprofit with no platform fee taken out.

The problem it solves

In the US, about 80 million pounds of food is wasted every single day. At the same time, 1 in 8 Americans does not have reliable access to enough food. Most of this waste happens because there is no easy way for a restaurant to quickly find someone who can take their leftovers before the food goes bad.

Current options are slow, require phone calls, or depend on personal connections. SaveFood makes the handoff fast, tracked, and visible to everyone involved.

Who is it for?

The app serves four different types of users, each with their own view and set of features.

- Restaurants that want to post leftover food for pickup instead of throwing it away.
- Nonprofits that collect, store, and distribute rescued food to people in need.
- Recipients who want to find free food nearby for themselves or their family.
- Donors who want to support a nonprofit with money and see exactly how it is used.

2. Goals

What success looks like

- A restaurant can post a food listing in under 2 minutes.
- A nonprofit can find, accept, and navigate to a pickup without leaving the app.
- A recipient can reserve a free meal and check in with a QR code or PIN.
- A donor can send money to a specific nonprofit and see a real-time donation meter update.
- All four user types can see their own impact, such as meals rescued and waste prevented.

What we are not trying to do

- This is not a food delivery service. Users come to pick up food themselves or a nonprofit sends a volunteer.
- This is not a grocery store or food marketplace. All food listed is free.
- This is not a general donation platform. Donors can only give to verified nonprofits on the app.

3. User Roles

Each role gets a completely separate dashboard and flow after the role selection screen.

Restaurant

A business that has leftover food at the end of a shift. They post listings and confirm handoffs when a nonprofit arrives.

What they can do

- Post a food listing with the name, quantity, category, allergens, diet type, and a photo.
- Choose how the food will be picked up: nonprofit comes to the restaurant, the restaurant drops food at a smart fridge, or community members pick up directly.
- Set a pickup deadline so the food does not stay listed past its freshness window.
- Track the status of a listing in real time, from posted to nonprofit en route to confirmed handoff.
- Confirm the handoff when the nonprofit arrives and mark the food as delivered.
- See a monthly impact summary showing total meals rescued and pounds of waste diverted.
- Receive push notifications when a nonprofit accepts a listing or is approaching.

Nonprofit

An organization that picks up food, stores it in smart fridges or warehouses, and distributes it to people in need.

What they can do

- Browse nearby food listings on a map or a list view, sorted by urgency.
- See the freshness score of each listing, which is calculated from IoT sensor data where available.
- Accept a listing and get turn-by-turn navigation directions to the pickup location.
- Mark arrival and confirm the handoff inside the app.
- Monitor smart fridges they manage, including temperature, humidity, capacity, and individual slot inventory.
- Get alerted when a fridge item is about to expire or when a fridge door is left open.
- View a donation and impact dashboard showing funds raised, meals rescued, and how money was spent.

Recipient

An individual looking for free food nearby. This could be someone facing food insecurity, a student, or anyone who wants to reduce waste.

What they can do

- See nearby food listings on a map or browse a list sorted by distance and urgency.
- View the food details including freshness, allergens, quantity, and serving size.
- Reserve a time slot to pick up the food. The slot holds for 30 minutes.
- Receive a QR code and a 4-digit PIN to use at pickup.
- Check in at pickup by showing the QR code or entering the PIN to a volunteer.
- Browse smart fridges nearby, see what is available inside each one, and reserve an item.
- See a personal history of meals picked up and waste prevented.

Donor

Someone who wants to support food rescue efforts with money. Donors can browse nearby nonprofits, see their impact metrics, and donate directly.

What they can do

- Browse nonprofits near them with details on meals rescued per month, number of beneficiaries, and current fundraising progress.
- View a breakdown of how each nonprofit spends donations, such as transport, fridge maintenance, and outreach.
- Choose a donation amount from preset options or enter a custom amount.
- Pay with Apple Pay, Venmo, or a debit or credit card through Stripe.
- Receive a tax-deductible receipt by email after donating.
- See the donation meter update in real time after their contribution.
- Zero platform fee. The full amount reaches the nonprofit.

4. Screens and Features

The prototype contains 22 screens. They are grouped below by flow.

Onboarding

Splash screen

- Shows the SaveFood logo, tagline, and a Get Started button.
- Returning users can tap "I already have an account" to go to the login screen.

Login screen

- Email or phone number and password fields.
- Sign in with Google or Apple as alternative options.
- Forgot password link and a create account option for new users.

Role selection screen

- After login, users pick one of the four roles: Restaurant, Nonprofit, Recipient, or Donor.
- The app shows a short description under each role so users know what they are choosing.
- Users agree to the Terms and Food Safety Policy before continuing.
- Users can switch their role later from the profile screen.

Restaurant flow

Restaurant dashboard

- Shows the restaurant name, verification status, and location.
- Summary cards for total meals rescued and pounds of waste reduced.
- A quick post button to create a new listing right from the dashboard.
- Active listings with their current status, freshness score, time remaining, and pickup mode.
- A monthly impact summary section at the bottom.
- Bottom navigation with Home, Post, Listings, Alerts, and Profile tabs.

Create listing screen

- Form fields for food name, quantity in pounds, number of people it serves, category, diet type, and allergens.
- Three pickup mode options: nonprofit pickup, smart fridge drop-off, or direct user pickup.
- A pickup deadline field and an optional photo upload.
- A tip that food listed within 30 minutes gets faster pickup, which is shown at the top of the form.

Listing status screen

- Shows a full timeline of a specific listing from posted to picked up.
- Displays the accepting nonprofit name, distance, and estimated arrival time.
- A call button and a Confirm Handoff button at the bottom.

Handoff confirmation screen

- A success screen shown after the restaurant confirms the handoff.
- Shows the weight handed over and how many people it will serve.
- Updates the restaurant impact counter.

Nonprofit flow

Nonprofit dashboard

- Three view tabs at the top: Map view, List view, and Fridges view.
- Map view shows food pins near the nonprofit with urgency markers.

- List view shows listings sorted by urgency with freshness scores and time remaining.
- Fridge view shows smart fridges the nonprofit manages with capacity and temperature status.
- Summary stats at the top showing meals rescued and donations raised this month.

Listing detail screen for nonprofits

- Full details on the food item including allergens, quantity, freshness score with time remaining, and distance.
- An Accept and Navigate button that opens the in-app navigation screen.

Navigation screen

- A simplified turn-by-turn map view showing the route to the pickup location.
- ETA, distance, and pickup deadline shown as cards.
- A Mark as Arrived button that triggers the handoff confirmation flow.

Smart fridge detail screen

- Live sensor readings including temperature, humidity, CO2 level, door status, and power supply.
- Inventory list showing each slot, the food item, how long ago it was added, and its freshness status.
- A warning when a slot contains an expired item that needs to be removed.

Impact dashboard

- Total donations raised and progress toward the monthly goal.
- Number of donors, meals rescued, pounds of food diverted, and beneficiaries served.
- A breakdown of how donations are spent across transport, fridge maintenance, storage, and outreach.

Recipient flow

Food discovery screen

- A map showing nearby food listings and smart fridges with distance markers.
- Quick filter badges at the top for free food and fridge options.
- Bottom navigation with Nearby, Browse, Reserved, History, and Profile tabs.

Food list screen

- A list of available food items and smart fridges sorted by distance.
- Each card shows freshness score, urgency, serving size, and availability status.

Food detail screen

- Full details on the food item including allergens, freshness report with IoT verification, temperature it is stored at, quantity, and distance.
- The offering nonprofit name, hours, and verification badge.
- A Reserve This Food button.

Reserve flow screen

- Shows available pickup time slots for the next 90 minutes.
- Slots past the best-before window are shown as unavailable.
- Reservation rules are displayed: the slot holds for 30 minutes, and no-shows affect the community score.

Pickup timer screen

- A countdown timer showing how long is left to pick up the reservation.
- The exact address and door location for pickup.
- A 4-digit pickup code and instructions on what to bring.
- An I am Here button to start the check-in flow.

QR check-in screen

- A QR code the user shows to the volunteer at pickup.
- A 4-digit PIN as an alternative if scanning is not possible.
- Geofence verification shown under the PIN.

Pickup success screen

- Confirms the pickup with the food name and source nonprofit.
- Shows how much waste the user prevented.
- A prompt to rate the pickup and a button to find more food.

Smart fridge user view

- Shows the fridge name, location, temperature, number of items available, and hours.
- Lists each available slot with the food item, weight, servings, and freshness.
- A Reserve button on each available slot.
- Expired items are marked as not recommended and cannot be reserved.
- Instructions on how to open the fridge locker using the QR code or OTP after reserving.

Donor flow

Donor discovery screen

- Platform-wide stats at the top showing total raised this month and meals funded.
- A list of nonprofits near the donor with their fundraising progress, donor count, and meals per month.
- A Donate Now button on each nonprofit card.

Nonprofit profile screen

- Full profile of the nonprofit including when it was established, location, number of smart fridges, and verification status.
- Impact metrics: meals rescued per month, beneficiaries served, food waste diverted.
- A donation progress bar showing current amount, goal, and number of donors.
- A breakdown of how donations are used across four categories.
- A Donate to This NGO button.

Donate flow screen

- Preset donation amounts of \$20, \$100, and \$200, each showing the equivalent in rescued meals.
- A custom amount field.
- Zero platform fee messaging shown clearly on the screen.
- Payment options: Apple Pay, Venmo, or debit and credit card through Stripe.
- Tax deductible receipt note and 501(c)(3) eligibility disclosure.

Donation success screen

- Confirms the donation amount, recipient nonprofit, platform fee of zero, and a transaction ID.
- Shows the updated donation meter with the donor contribution counted in.
- A return to nonprofits button.

Shared screens

Notifications screen

- Shows alerts for all role-relevant activity: nonprofit en route, unclaimed listings, successful pickups, and new donations received.
- Each notification shows a timestamp and a short description.

Profile screen

- Shows the user name, role, and verification status.
- Links to listings, impact report, notification settings, verification status, and help.
- Total impact summary showing meals rescued and waste saved.
- A Switch Role option and a Log Out button.
- A navigation map at the bottom showing all 22 prototype screens grouped by role, for demo purposes.

5. Smart Fridge and IoT Features

Smart fridges are a key part of the SaveFood system. They let restaurants drop off food without needing to coordinate a live pickup, and they give recipients 24-hour access to food in public locations.

What the fridge tracks

- Temperature, shown in Celsius with an ideal range of 2 to 5 degrees.
- Humidity, shown as a percentage.
- CO2 and air quality inside the fridge.
- Door status: open or closed, and how long since it was last opened.
- Power supply status, including any outage alerts.
- Each slot in the fridge, with the food item name, when it was added, weight, and freshness level.

Freshness scoring

Each food item in the fridge has a freshness score shown as a percentage. The score is calculated from sensor data and the time since the food was added. Items near expiry show a warning, and expired items are marked as not recommended so recipients cannot reserve them.

Access control

When a recipient or nonprofit reserves a fridge slot, they receive a QR code and a numeric OTP. When they arrive at the fridge and scan or enter their code, the correct locker opens automatically. This means no volunteer or staff member needs to be present at the fridge location.

6. Freshness Score

The freshness score appears across multiple screens and is one of the more important features of the app. It gives all users a simple signal about how safe and good the food is.

- Scores from 90 to 100 are labeled Excellent.
- Scores from 70 to 89 are labeled Good.
- Scores from 50 to 69 are labeled Consume Soon.
- Scores below 50 trigger a Remove or Not Recommended warning.

For food stored in a smart fridge, the score is calculated using live IoT data. For food posted directly by a restaurant, the score is estimated based on when the food was prepared and the current time.

7. Notifications

The app sends push notifications to users based on their role and the status of their active listings or reservations.

Restaurant notifications

- A nonprofit has accepted your listing.
- A nonprofit is on the way and is estimated to arrive in a certain number of minutes.
- A listing is about to expire and has not been claimed.
- A food pickup was confirmed successfully.
- A donation was received by a nonprofit partner.

Nonprofit notifications

- A new urgent listing is available near you.
- A fridge slot is expiring soon.
- A fridge door has been open for an unusual amount of time.
- A new donation has come in.

Recipient notifications

- Your reservation is about to expire.
- A food item near you is about to expire and is still available.

8. Rules and Edge Cases

Reservations

- A reservation holds the food slot for 30 minutes from the selected pickup time.
- If the recipient does not check in within the 30-minute window, the reservation is automatically released and the food becomes available to others.
- Repeated no-shows lower the recipient community score.
- Only pickup time slots before the food best-before window can be selected. Slots that fall after the best-before time are shown as unavailable.

Listings

- A food listing is visible to nearby nonprofits and recipients as soon as it is posted.
- If no one claims a listing and the pickup deadline passes, the restaurant gets a notification that the listing is unclaimed.
- A listing can only be confirmed as handed off by the restaurant after the nonprofit or recipient physically arrives.

Donations

- The platform takes zero fees from donations. The full amount goes to the nonprofit.
- Donors receive a tax-deductible receipt by email after each donation.
- Donations are processed through Stripe.

Verification

- Restaurants and nonprofits must go through a verification process before they can post listings or accept food.
- Verified status is shown with a badge on their profile and on their cards in the app.
- Unverified users can browse but cannot post or accept food.

9. What Is Not Built Yet

This section covers features that are shown in the prototype as static or implied but are not yet implemented as real functionality.

- Real-time GPS tracking of nonprofit vehicles en route to a pickup.
- Live IoT sensor integration with actual fridge hardware.
- Backend user accounts, login, and authentication.
- Real payment processing through Stripe.
- Push notification delivery infrastructure.
- Community score system for recipients.
- Actual QR code generation and scanning.

- Geofence verification at fridge locations.
 - Nonprofit and restaurant verification workflow.
 - Search and filtering for listings by food type, allergen, or distance.
 - Admin dashboard for platform oversight.
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10. Open Questions

These are decisions that need to be made before building the real product.

- How does freshness scoring work for food that is not inside a smart fridge? Is the score purely time-based or does the restaurant have to enter preparation time manually?
 - What happens to a listing if the restaurant closes before a nonprofit arrives?
 - How do we handle allergic reactions or food safety complaints? Is there a reporting flow?
 - Who owns and maintains the smart fridges? Is that the nonprofit, the platform, or a third party?
 - Is the community score for recipients public or only visible to the recipient themselves?
 - Does the app support multiple restaurant locations under one account?
 - What language options does the app need to support? The prototype is English-only.
 - Is there a cap on how much food a recipient can reserve per day or per week?
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Appendix: Screen List

All 22 screens in the prototype, grouped by role.

Onboarding (3 screens)

- Splash
- Login
- Role Selection

Restaurant (5 screens)

- Restaurant Dashboard
- Create Listing
- Listing Status
- Handoff Confirmation
- Notifications

Nonprofit (5 screens)

- NGO Dashboard with Map, List, and Fridge views
- Listing Detail and Accept
- Navigation
- Smart Fridge Detail
- Impact and Donations Dashboard

Recipient (7 screens)

- Food Discovery Map
- Food List
- Food Detail
- Reserve Flow

- Pickup Timer
- QR Check-in
- Pickup Success
- Smart Fridge User View

Donor (4 screens)

- Donor Discovery
- NGO Profile
- Donate Flow
- Donation Success

Shared (2 screens)

- Profile
- Notifications